Reliability Engineering and System Safety 113 (2013) 21–29

Contents lists available at SciVerse ScienceDirect

Reliability Engineering and System Safety

journal homepage: www.elsevier.com/locate/ress

Optimal maintenance strategy of deteriorating system under imperfect maintenance and inspection using mixed inspection scheduling

Minh Duc Le*, Cher Ming Tan [1]

School of Electrical Electronics Engineering, Nanyang Technological University, 50 Nanyang Avenue, Singapore 639798, Singapore

ARTICLE INFO

Article history:
Received 28 August 2011
Received in revised form
22 June 2012
Accepted 12 November 2012
Available online 12 December 2012

Keywords:
Multi-state system
Markov process
Imperfect maintenance
Imperfect inspection
Sequential inspection
Continuous monitoring

ABSTRACT

An operating system suffers from degradation. Each degradation level can be represented by a state, making the system a multi-state system. In many situations, there are no apparent symptoms indicating the system's state, and system's degradation level can only be known through thorough inspection. Through condition monitoring, the system's state can be estimated with some uncertainty. In this work, we investigated inspection-maintenance schemes for such a system. Our assumptions are that the maintenance is imperfect and the degradation is a continuous-time Markov process. We proposed a strategy that combines both inspection and continuous monitoring to reduce unnecessary thorough inspection and to improve the system's reliability. Optimal maintenance strategy is derived based on an iterative algorithm to minimize the mean long-run cost rate.

1. Introduction

Operating systems suffer from degradation. The degradation levels can often be indexed by various measurable physical quantities. When the degradation index reaches a certain critical level, the system does not function satisfactorily and need to be replaced or maintained. A number of maintenance policies have been proposed. These policies span from the most basic one as corrective maintenance (CM) to more advanced policy as preventive maintenance (PM) [1], and to the recent condition-based maintenance (CBM) [2].

A corrective maintenance policy is an unscheduled maintenance and is carried out only when the system fails. This approach is hardly used in the maintenance of highly complex systems. Preventive maintenance policy, on the other hand, is proactively designed to prevent the failure from happening by maintaining the system before it reaches a pre-defined degradation level. Conventionally, PM policy often involves periodical replacement, block maintenance, or pre-defined time of operation based solely on system's failure history. While being effective for the maintenance of single-unit systems, traditional PM does not prevail for multi-unit systems due to the dependence between units. A minor failure of one unit may lead to different behaviors of the others in the same system, which requires correspondent changes in the PM policy. Such systems should be considered as multi-state systems (MSS) and need different treatments. Condition-based maintenance (CBM) is proposed in order to further improve PM policy for multi-state systems.

A CBM program determines maintenance decisions based on the actual system's health condition thus extends the system's lifecycle and lowers the maintenance costs significantly [3,4]. CBM, by its definition, is majorly concerned with the observation of the system's health, and hence, estimating the system's condition is crucial to the CBM program's effectiveness. There are two ways to estimate the system's condition: thorough inspection and continuous monitoring. With the advancement of sensor technology, the system's health condition can be observed continuously. Moustafa et al. [5] developed a CBM policy for continuous monitoring, in which two maintenance options are considered, namely replacement and minimal repair. Although continuous degradation is commonly used in detecting system's degradation, it usually swarms with unnecessary and excessive data [4]. Besides, the system's state is only partially revealed or subjected to many diagnosis errors: measurement errors, false alarm etc, especially in complex systems which require huge number of monitoring devices [6]. There are also many works on the policy when maintenance is only brought fought upon sequential inspection result. Under these policies, inspections are carried out at designed time during operation. A good summary can be found in [2,7–9]. In the formulation of such polices, the comprehensive cost analysis showed that an optimal choice of inspection

* Correspondig author. Tel.: +65 8419 6731.
E-mail addresses: Lemi0006@e.ntu.edu.sg (M.D. Le), ecmtan@ntu.edu.sg (C.M. Tan).
[1] Tel.: +65 6790 4567.

0951-8320/$ - see front matter © 2012 Elsevier Ltd. All rights reserved.
http://dx.doi.org/10.1016/j.ress.2012.11.025

date and replacement threshold can improve the cost effectiveness of the maintenance policy.

In reality, some latent failures of a unit in the system are neither always obvious nor easy to detect. In some complex system, it is not practical to employ sensors to every unit, thus making its failure hidden to the monitor [10]. It can also happens that the monitoring system fails and delivers incorrect measurement, thus results in false alarms or missing of failure [11,12]. Besides employing sensor network for continuous monitoring, there are more accurate inspection activities such as vibration analysis, oil analysis, infrared and acoustic scan and others nondestructive testing. However these tests take time and must be carried out offline, thus increase the cost of maintenance. This type of inspection will be ineffective if not scheduled timely. The comparison of condition monitoring and sequential inspection was done by Sherwin and Lam et al [13,14].

In practice, it is unrealistic to consider the system under either continuous monitoring or inspection. A complex industrial system is often subjected to the combination of many monitoring devices as well as inspection schedules. This paper provides a framework for the analysis of such a combination and develops an algorithm to ensure timely maintenance action.

It is widely assumed that the imperfect maintenance restores a system to a state between as good as new (replacement) and as bad as old (minimal repair). The two extreme cases are investigated thoroughly in early works. The perfect replacement assumption is only reasonable for systems with few components which are structurally simple. In contrast, the minimal repair assumption is only applicable for failure of highly complex systems when only few of its components are replaced [15]. In general, these assumptions are not true in many applications. Furthermore, imperfection can arise during maintenance due to the maintenance engineering skills, quality of the redundant parts and complexity of the degraded systems. In order to implement a CBM policy practically, it is necessary to have a model that can predict the future degradation and failure of the system under such imperfections. Several theoretical models are developed that taking into account the imperfect maintenance. They can be broadly classified into several classes. The first class uses the concept of *virtual age* first proposed by Kijima et al. [16,17] in which the system's effective age is considered to be lower as a result of the repair. This approach has been intensively used due to its simplicity in determining the optimal maintenance interval. Its extension includes PAS and PAR models [18–20]. The second class is the *improvement factor* method in which either the failure rate [21,22] or the system's reliability [4,23,24] is improved after repair. However, these two approaches do not accounted for the actual system's condition and should be limited for time-based maintenance, rendering them unsuitable for CBM modeling. The third class is the *probabilistic approach* where it is assumed that the system undergoes "perfect renewal" to "as-good-as-new" condition with a constant probability of p and "minimal repair" to "as-bad-as-old" condition with a probability of $(1-p)$ [24–26]. This method is well developed in [27] as the author considered the number of imperfect maintenance conducted in the previous circle. The final class is based on the *system degradation* in which the system suffers random shocks at variable intervals of time causing it to undergo progressive increments of damage [19]. The effect of the imperfect maintenance actions is described by the degree of reduction of the cumulative system damage after repair as compared to that before repair. For detailed discussion on the various models, one can refer to Pham and Wang [15,28], Brown and Proschan [29] and Liu [30]. In this paper, we combine the last two approaches to form a probabilistic imperfect maintenance model for multistate systems.

The novelty of this work lies in determining the modeling of a multi-state system, taking imperfect maintenance and inspection into consideration. Herein, each state represents a level of degradation which can be detected accurately through inspection or partially by continuous monitoring. Due to deterioration, the system can only transit to higher degradation level, leading to poorer performance. Maintenance is applied to bring the system to a probably better state (lower degradation level). At each time a system state is detected by inspection, and decision must be made based on the system's detected condition. The decision can be either to replace the degraded system by an as-good-as-new one, to maintain it to a less degraded level, or to continue operating. By considering the cost of inspection, operation, and maintenance, an algorithm is proposed to optimize the policy so that the mean long-run cost rate is minimized.

2. System description and maintenance and inspection model

The system under study is a multi-state system, and each state represents a system's health condition. These states can be defined by either a degradation index such as vibration's intensity, temperature, etc, or simply the system's performance. The system is assumed to take a finite number of states 1,2,3,…,N where state 1 is the as-good-as-new state and state N is the completely failed state. The states are in ascending deteriorating order.

The degradation process is represented by the transition from one state to another state. The failures of a complex system have been shown to follow the exponential distribution despite the fact that the individual components in the system may follow different distribution [31]. Hence, the system's deterioration process can be modeled as a continuous-time Markov process. From state i, $(1 \leq i \leq N-1)$ the system can only transit to the more degraded state j, $(i \leq j \leq N)$ with a transition rate of λ_{ij}. In this work, for the sake of simplicity, we assume that the transition rates are constant for a given QUOTE and j. In actual cases, the transition rate can be changed after the system is maintained, and such cases will be considered in our future work.

Due to the lack of sensor's accessibility to some part of the system and the nature of hidden failures, continuous monitoring only partially detects the state of the system. The monitoring devices can reveal the range of states in which the system is operating; however the exact state is uncertain. These operating ranges can be denoted as M_1, M_2, M_3,…,M_m as illustrated in Fig. 1. An example can be the degradation of electric cars' battery, where continuous monitoring of the battery's operation can only detect if the battery is degraded. On the other hand, with the employment of particular inspections (vibration analysis, oil analysis, infrared and acoustic scan and others nondestructive testing), the exact degradation state is uncovered so that a more effective and specific maintenance actions

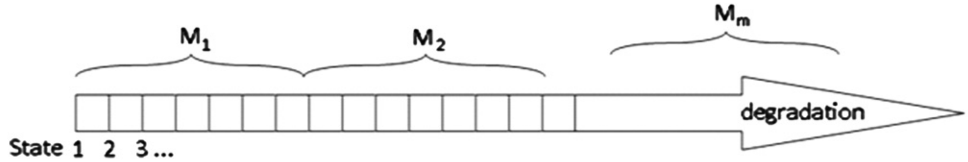

Fig. 1. Degradation levels that can be detected by continuous monitoring system.

can be carried out. Although such inspection is accurate, it is costly and cannot give a timely warning of a possible failure. In this work, we refer this type of inspection as sequential inspection. Sequential inspections will be used in combination with continuous monitoring system to enhance the system's reliability and effectiveness. In our proposed inspection policy, the system is always under continuous monitoring and a sequential inspection is carried out at planned time or when the continuous monitoring system indicates a change in the operation range. After that, based on the indicated state i, the system is either replaced, maintained or leave operating. If the system is left to operate without any action, a planned waiting time interval t_i needs to be determined for the next inspection.

The cost for each successive sequential inspection is fixed at C^{SI}. The system's failure (system at state N) is detected without inspection and not recoverable by repair. The system upon replacement is recovered to the initial state QUOTE with a cost of C^R. The failure also results in a consequential damage such as unplanned delay in production, lost of physical assets etc, which is represented by a cost of C^f.

The maintenance is assumed to be imperfect. Upon maintenance, instead of recovering to as-good-as-new condition, the system's state is improved to some better state j, $i \ge j \ge 1$, with probability P_{ij}^M. This probability can be obtained from the maintenance archive, which often contains the state of degradation QUOTE before maintenance and the improved state j after maintenance. The improved state j after maintenance often varies due to the uncertainty during maintenance work. With a sufficient maintenance data, the maintenance probability can be estimated. Generally, the probability that the system is recovered to as-good-as-new state is getting smaller as the system state is approaching N, and the maintenance cost and time varies with different states. C_i^M denotes the mean cost of maintenance to repair the system at state i. For the sack of simplicity, the mean maintenance cost at state i is averaged on the cost to maintain to any the improved state j: C_{ij}^M, so that it only depends on the current degradation state $C_i^M = \sum_j P_{ij}^M C_{ij}^M$. The maintenance cost also includes the cost due to system unavailability.

The cost of operation is increasing with the degradation in order to accounts for the loss in profit due to the degradation in the system's performance. The operation cost rate when the system is operating at state i is denoted as a_i.

3. Mathematical formulation of inspection-maintenance scheduling

The system is put under continuous inspection at all time. Whenever the continuous monitoring indicates a change in the operation range, sequential inspection is carried out to detect the exact system state. After the system's current state $i \ne N$ is indicated by sequential inspection, the maintenance and inspection policy determine the decision to be made. The decision can be to replace, to maintain or to leave the system in continuous operation and do no maintenance action. If the system is left to operate without any action, a planned waiting time interval t_i needs to be determined for the next inspection. Sequential inspection will be carried out after this waiting time t_i is elapsed or a change in operation range is detected by continuous monitoring.

To determine the optimal maintenance policies, let us define the following terms:

- δ: A policy which determines the action at each state, either replacement, maintenance or leave the system in continuous operation.
- $X_\delta(i)$: Mean operating time from the moment the system is detected by sequential inspection to be at state i to the time of system replacement.
- $Y_\delta(i)$: Mean cost from the moment the system is detected by sequential inspection to be at state i to the time of system replacement.
- $D(i)$: Decision at state i. It can be either to inspect the system after time interval t_i, denoted as $I(t_i)$, to maintain, denoted as M, or to replace, denoted as R.

If the system fails, it is automatically detected and must be replaced. The cost of replacement and the induced cost of failure is $Y_\delta(N) = C^R + C^f$.

We have the following situations:

a. If the decision is to replace $(D(i)=R)$
Upon replacement, the system is directly recovered to state 1. We have

$$X_\delta(i) = 0 \tag{1}$$

$$Y_\delta(i) = C^R \tag{2}$$

If the decision is to maintain $(D(i) = M)$
Upon maintenance, the system at state i is recovered to state j with a probability P_{ij}^M of Hence we have

$$X_\delta(i) = \sum_{j=1}^{N} P_{ij}^M X_\delta(j) \tag{3}$$

$$Y_\delta(i) = C_i^M + \sum_{j=1}^{N} P_{ij}^M Y_\delta(j) \tag{4}$$

b. If the decision is to leave the system to continuous operation until next inspection or a change in the operation range is detected by continuous monitoring, $(D(i) = I(t_i))$

Given that the system is initially at state i, we define the following:

- $s(u)$ is the state of the system after the interval QUOTE from the time the system is detected to be in state i
- $R(i)$ is the operating range contains the state i, i.e. one of the range M_k as depicted in Fig. 1
- $P_i(s(t_i)=j)$ is the probability that the system is detected to be at state j after an interval t_i
- $P_i(s(u) \in R(i))$ is the probability that the system's state $s(u)$ after an interval u is in the operation range containing state i, which is also the probability that no change in operation range is detected by continuous monitoring after an interval u. This probability can be calculated as $P_i(s(u) \in R(i)) = \sum_{j \in R(i)} P_i(s(u) = j)$
- $A_i(t)$ is the mean operating cost from the time the system is evaluated to be at state i until time t. Hence, $A_i(t_i)$ is the mean operating cost from the time the system is detected to be at state i until the next planned inspection. QUOTE can be expressed as:. $A_i(t) = \int_0^t \sum_{j \in R(i)} P_i(s(u) = j) a_j du$

We have

$$X_\delta(i) = \int_0^{t_i} P_i(s(u) \in R(i)) du + \sum_{j=1}^{N} P_i(s(t_i) = j) X_\delta(j) \tag{5}$$

$$Y_\delta(i) = A_i(t_i) + C^{SI} P_i(s(u) \neq N) + \sum_{j=1}^{N} P_i(s(t_i)=j) Y_\delta(j) \tag{6}$$

The derivation of Eqs. (5) and (6) are shown in the Appendix.

The probability $P_i(s(t)=j)$ can be calculated by solving the Kolmogorov forward equations [32]. In these equations, the transition rates between states that are not in the same operating range with the state i will not be accounted since any change of state out of the current operating range will be detected and interrupted by responsive maintenance actions. Given that the system is initially at state i, the probability that the system is detected to be in the state j after time interval t is governed by:

$$\frac{d}{dt} P_i(s(t)=j) = -\sum_{k=1,\, j \in R(i)}^{N} \lambda_{jk} P_i(s(t)=j) + \sum_{k \in R(i)} \lambda_{kj} P_i(s(t)=k) \tag{7}$$

In Eq. (7), the first term $-\sum_{k=1,\, j \in R(i)}^{N} \lambda_{jk} P_i(s(t)=j)$ represents the rate that the system transits from state j to any state K. State j must be in the same operating range as state i for this rate to be counted. The second term $\sum_{k \in R(i)} \lambda_{kj} P_i(s(t)=k)$ represents the rate that the system transits from any state K to state j. State K must be in the same operating range as state i for this rate to be counted.

The probability $P_i(s(t)=j)$ can be calculated by solving the following equation.

$$\frac{d}{dt}\begin{bmatrix} P_i(s(t)=1) \\ P_i(s(t)=2) \\ \vdots \\ P_i(s(t)=N) \end{bmatrix} = Q(i) \begin{bmatrix} P_i(s(t)=1) \\ P_i(s(t)=2) \\ \vdots \\ P_i(s(t)=N) \end{bmatrix} \tag{8}$$

where $Q(i)$is the transition matrix and its components are given as

$$Q_{jk}(i) = (k \neq j \;\&\; k \in R(i)) \times \lambda_{kj} - (k=j \;\&\; j \in R(i)) \times \sum_{l=1}^{N} \lambda_{jl}$$

In the expression of $Q_{jk}(i)$ in Eq. (8), $(k \neq j \;\&\; k \in R(i))$ and $(k=j \;\&\; j \in R(i))$ are the logic function, taking values of 0 or 1, $(k \neq j \;\&\; k \in R(i)) \times \lambda_{kj}$ represents the transition rate that the system transits from state $k \in R(i)$ to another state j if $j \neq k$, and $(k=j \;\&\; j \in R(i)) \times \sum_{l=1}^{N} \lambda_{jl}$ represents the transition rate that the system transits from state $j \in R(i)$ to others state l if $j=k$.

Overall, we have:

$$X_\delta(i) = \begin{cases} 0, \textit{ if } D(i)=R \\ \sum_{j=1}^{N} P_{ij}^{M} X_\delta(j), \quad \text{if} \quad D(i)=M \\ \int_0^{t_i} P_i(s(u) \in R(i))du + \sum_{j=1}^{N} P_i(s(t_i)=j) X_\delta(j), \quad \text{if} \quad D(i)=I(t_i) \end{cases} \tag{9}$$

$$Y_\delta(i) = \begin{cases} C^R, \textit{ if } D(i)=R \\ C_i^M + \sum_{j=1}^{N} P_{ij}^{M} Y_\delta(j), \quad \text{if} \quad D(i)=M \\ A_i(t_i) + C^{SI} P_i(s(u) \neq N) + \sum_{j=1}^{N} P_i(s(t_i)=j) Y_\delta(j), \quad \text{if} \quad D(i)=I(t_i) \end{cases} \tag{10}$$

The optimal policy must minimize the overall mean cost rate over the system's life cycle. Hence the overall mean cost rate can be calculated as $g = Y_\delta(1)/X_\delta(1)$, where $X_\delta(1)$ and $Y_\delta(1)$ are the mean operating time and cost of the system from new (state 1) to replacement.

In order to find the optimal policy δ^*, we define $V_\delta(i) = Y_\delta(i) - gX_\delta(i)$. The optimal policy must maximize the operating time $X_\delta(1)$ and minimize the mean overall cost $Y_\delta(1)$, i.e. to minimize $V_\delta(i) = Y_\delta(i) - gX_\delta(i)$. The optimal policy δ^* and the minimal cost rate g^* can be obtained by solving the following set of equation:

$$V_\delta(1) = Y_\delta(1) - g^* X_\delta(1) = 0 \tag{11}$$

$$V_\delta(N) = Y_\delta(N) - g^* X_\delta(N) = C^R + C^f \tag{12}$$

$$V_\delta(i) = \min \begin{cases} C^R & \text{for} \quad D(i)=R \\ C_i^M + \sum_{j=1}^{N} P_{ij}^{M} V_\delta(j) & \text{for} \quad D(i)=M \\ \min_{t_i} \left[A_i(t_i) + C^{SI} P_i(s(u) \neq N) - g^* \int_0^{t_i} P_i(s(u) \in R(i))du + \sum_{j=1}^{N} P_i(s(t_i)=j) V_\delta(j) \right] & \text{for} \quad D(i)=I(t_i) \end{cases} \tag{13}$$

Eqs. (11)–(13) can be iteratively solved by applying the following algorithms:

- Step 1
  - Set $V_\delta(i) = C^R + C^f, \forall\, i$
  - Set $g = g_0$ as an initial value
- Step 2
  - For $i = 1{:}(N-1)$ Calculate $V_\delta(i)$ using Eq. (13)
- Step 3
  - Repeat step 2 until the value of $V_\delta(i)$ converge
- Step 4
  - If $V_\delta(1) \neq 0$, update g using Newton–Raphson method

  $$g_n = g_{n-1} + \frac{V_\delta(1)}{\sum_{D(i)=I(t_i)} \int_0^{t_i} P_i(s(u) \in R(i))du}$$

  - If $V_\delta(1) = 0$, . $g^* = g$, $\delta^* = \delta$ Stop

This system model can be specified to cover some special cases described in the literature as follows.

1. Corrective Maintenance:
   In this case, there's no continuous monitoring involved, hence the system can only be monitored as "operating" or "failed" $R(i) = \{1,2,\ldots,N-1\}$ for $i \neq N$ and the system is left operating until breakdown $(D(i) = I(\infty)\ \forall i < N)$
2. Sequential Inspection Policy [8]:
   In this case, there's no continuous monitoring involved. Hence, between inspections, the system can only be monitored as "operating" or "failed" $(R(i) = \{1,2,\ldots,N-1\}$ for $i \neq N)$. Upon

inspections, the system is maintained, replaced or left operating until next planned inspection. In [8], maintenance is assumed perfect such that any maintenance action can renew the whole system hence $P^M_{i1}=1\ \forall i$.

3. Continuous Inspection Policy [9]:
The monitoring system is assumed perfect such that any changes in system states are detectable ($R(i)=\{i\}, \forall i$). Besides, maintenance is also perfect ($P^M_{i,1}=1, \forall i$). Maintenance actions can be carried out timely once the system exceeds its maintenance threshold.
4. Continuous Inspection with Minimal Repair [5]:
This case is similar to the case 3, however, minimal repair is considered together with replacement. Minimal repair effect is modeled as an improvement of one state ($P^M_{i,i-1}=1, \forall i$). The system is either under minimal repair or replacement once exceeding its maintenance threshold.

4. Numerical results and discussion

4.1. Description of the case study

In this section, a hypothetical system is simulated to illustrate the impact of different policies on the system's total cost and number of maintenance. The system consists of 13 states, which represents the system degradation levels in ascending order. State 1 is the state of no degradation (best performance) and state 13 is the state of total failure (worst performance). At any time during operation, the system can be degraded to a higher state or experienced a shock so that it fails immediately. A random shock may cause an instant failure to the system hence bring it directly to state N. Meanwhile the degradation process gradually lower the system's performance as it transits to the next state ($i\rightarrow i+1$). Besides, the system may undergo some minor failures that cause it to severely degrade to a much higher state. Due to the fact that the more degraded the system is, the higher probability that it become more degraded and fails, hence it is assumed that the degradation rate and the failure rate increase with the system state. The inter-state transition rate and the system's failure rate at each state are assumed and given in Table 1. In Table 1, $\lambda_{ij}=NA$ means no possible transition from state i to state j.

Due to the loss caused by degradation, namely lower productivity and higher resources consumption, the cost of operation is increasing with the degradation levels. The maintenance cost must account for the loss of availability due to maintenance time; hence the maintenance cost is also increasing with the degradation level. The operation, replacement and maintenance cost for different states are assumed and given in Table 2. The cost for each sequential inspection is $C^{SI}=1.0$ (,000 \$)

The continuous monitoring system is able to monitor 4 ranges of operating. These ranges are the states: ($1\rightarrow 3$), ($4\rightarrow 9$), ($10\rightarrow 12$), and the state of failure (13). These ranges are correspondingly denoted as 'good', 'average', 'bad' and 'failed' as shown in Table 3.

Table 4 gives the conditional probability of the post-maintained j state given the pre-maintained state i. For example, given that the state before maintenance is 3, the probability that the system will recover to state 2 will be 0.18. For this case study, these values are calculated based on a hypothetical maintenance model. Our system consists of n identical sub-systems in parallel, and state i of the system is correspond to i number of the sub-systems have failed. Each time maintenance is carried out, all the failed sub-systems will have equal probability p to be recovered. Hence, the probability of post-maintained state j is the probability that $i-j$ sub-systems are recovered. As the sub-system is either recovered or remain failed, we can consider the probability of post-maintained state to be at state j from state$i$ as a binomial distribution given as $P^M_{ij}=\binom{i-1}{j-1}(1-p)^{j-1}p^{i-j}$. It follows that the expected post-maintained state and its variance are both linearly increasing with the pre-maintained state i, i.e. $E(j)=p+(1-p)i$ and $var(j)=(i-1)p(1-p)$. These indices indicate that as the system is more degraded, it is more difficult to maintain the system to the initial condition and the consistence of the maintenance quality decreases. In this case study, the sub-system recovery probability is arbitrary assumed to be $p=90\%$. For this value of p, $P^M_{32}=\binom{3-1}{2-1}(1-0.9)^{3-1}0.9^{3-2}=0.18$.

4.2. Optimal maintenance policy

Without any maintenance policy applied, the given system has state probability as shown in Fig. 2. Fig. 2 is obtained by solving

Table 1
System's inter-state transition rate λ_{ij} (month^{-1}).

State	$J=1$	2	3	4	5	6	7	8	9	10	11	12	13
$i=1$	NA	.47	NA	.05	NA	NA	.002	NA	NA	NA	NA	NA	.0002
2	NA	NA	.48	NA	.05	NA	NA	.003	NA	NA	NA	NA	.0005
3	NA	NA	NA	.50	NA	.06	NA	NA	.003	NA	NA	NA	.001
4	NA	NA	NA	NA	.51	NA	.06	NA	NA	.004	NA	NA	.002
5	NA	NA	NA	NA	NA	.53	NA	.06	NA	NA	.004	NA	.004
6	NA	NA	NA	NA	NA	NA	.55	NA	.07	NA	NA	.005	.009
7	NA	NA	NA	NA	NA	NA	NA	.57	NA	.07	NA	NA	.01
8	NA	NA	NA	NA	NA	NA	NA	NA	.58	NA	.07	NA	.03
9	NA	NA	NA	NA	NA	NA	NA	NA	NA	.60	NA	.08	.07
10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	.62	NA	.15
11	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	.64	.31
12	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	.65
13	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

Table 2
Maintenance-operation cost description (000 \$).

State	$i=1$	2	3	4	5	6	7	8	9	10	11	12	13
C^M_i		5.0	5.0	5.0	5.1	5.1	5.2	5.2	5.3	5.4	5.6	6.2	$C^R=6.5$
a_i	0.82	0.86	0.91	0.95	1.00	1.05	1.11	1.16	1.22	1.29	1.35	1.32	
C^f													50.0

the set of Eq. (7) under the assumption that neither condition monitoring nor maintenance is applied. State 13 represents the state of failure; its corresponding probability is the cumulative density function of the failure distribution of the system. Taking integration of the reliability function over the entire life-cycle of the system, i.e.$\int_0^{\infty} \overline{F}(t)dt$, we have the system's mean time to failure of approximately $\tau_{MTTF} \sim 12$ months.

To compare the effectiveness of different maintenance-inspection policies with the optimal policy, some policies are proposed and Monte Carlo simulations are run for all these policies to compare the system's mean overall cost. We first investigate some continuous inspection policies with a maintenance threshold as shown in Table 5. There are 12 degradation states corresponding to 12 possible continuous inspection policies. Herein, we only investigate 7 policies; P1, P2, P3, P4, P5, P7, P9; with the ascending maintenance thresholds as states 1,2,3,4,5,7,9. If the state exceeds the policy's threshold, the system is maintained or replaced. Inspection is not planned ($I(\infty)$) and it is carried out only after a change in operating range is detected by

Table 3
Observable operating ranges.

State	$i=1$	2	3	4	5	6	7	8	9	10	11	12	13
$R(i)$	'GOOD'				'AVERAGE'						'BAD'		'FAILED'

Table 4
Maintenance probability P^M_{ij} from State $i \to j$.

State	$j=1$	2	3	4	5	6	7	8	9	10	11	12	13
$i=1$													
2	.9	.1											
3	.81	.18	.01										
4	.73	.24	.027	.001									
5	.65	.29	.048	.004	.0001								
6	.59	.32	.072	.008	.0004								
7	.53	.35	.098	.015	.0012	.0001							
8	.47	.37	.12	.023	.0026	.0002							
9	.43	.38	.14	.033	.0046	.0004							
10	.38	.38	.17	.044	.0074	.0008							
11	.34	.38	.19	.057	.011	.0015	.0001						
12	.31	.38	.21	.071	.015	.0025	.0003	0.0001					
13													

the monitoring system or maintenance. For example, for the policy P4, with the maintenance threshold is at state 5. The system starts from new (state 1) and is left to operate until it degrades to a state out of the current operating range according to Table 3. The exact degradation state is assessed by sequential inspection. If the state inspected is found to be at state 6 which exceed the threshold, the system must be maintained until the post-maintained state is below the maintenance threshold. It is then left to continue operating.

Employing the iterative algorithms in Section 3, we obtain the optimal policy for the described system. The optimal maintenance policy (P.Optimal) is described in Table 6. Under P.Optimal, the first three states (1, 2, 3) are left operating without inspection interfering ($I(\infty)$), until the monitoring system detect a change in operating range. State 4 will be re-inspected after $t_4=4$ months of operating ($I(4)$). State 5 and 6 will be maintained and the higher states will be treated by replacement. Two policies P4.under and P4.over are introduced to compare the effect of the time t_i between inspections. In the policies P4.under and P4.over, the planned waiting times are respectively $t_4=6$ and $t_4=2$, comparing to the optimum $t_4=4$. They can be interpreted as the over-inspected and under-inspected policies.

4.3. Simulation of system states and total cost

The behavior of the system changes with different policies. Monte Carlo simulations are run for the ten described policies for an operating time of 36 months. Their mean number of failure, maintenance and total cost are recorded as shown in Table 7. The system mean time to failure is ~ 12 months as computed earlier, thus the mean number of failure under corrective maintenance is approximately 3 for 36 months of operation. In Table 7, it is obvious that all the maintenance policies studied here significantly reduce the mean number of failure to less than 0.5, indicating their effectiveness as compared to the corrective maintenance.

To examine the effect of maintenance threshold on the mean overall cost rate, we have the cost rate associated with each policy as shown in Fig. 3, and, one can see that the mean cost rate has a minimum at Policy P4. P1 to P3 are over maintained since the mean overall cost rate decrease with the rate of maintenance. The maintenance rate in P5, P7, and P9 are low, resulting in high number of failure, and thus the mean overall cost rate increases. It

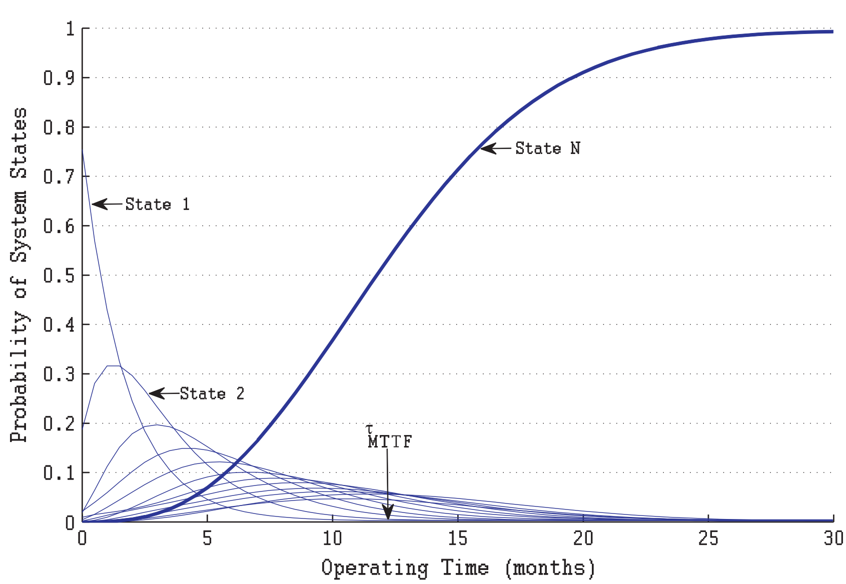

Fig 2. Probability of system state versus operating time under corrective maintenance policy.

Table 5
Inspection-maintenance policies under simulation.

Policy	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	S13
P1	$I(\infty)$	M	M	M	M	M	R	R	R	R	R	R	R
P2	$I(\infty)$	$I(\infty)$	M	M	M	M	R	R	R	R	R	R	R
P3	$I(\infty)$	$I(\infty)$	$I(\infty)$	M	M	M	R	R	R	R	R	R	R
P4	$I(\infty)$	$I(\infty)$	$I(\infty)$	$I(\infty)$	M	M	R	R	R	R	R	R	R
P5	$I(\infty)$	$I(\infty)$	$I(\infty)$	$I(\infty)$	$I(\infty)$	M	R	R	R	R	R	R	R
P7	$I(\infty)$	$I(\infty)$	$I(\infty)$	$I(\infty)$	$I(\infty)$	$I(\infty)$	$I(\infty)$	R	R	R	R	R	R
P9	$I(\infty)$	$I(\infty)$	$I(\infty)$	$I(\infty)$	$I(\infty)$	$I(\infty)$	$I(\infty)$	$I(\infty)$	$I(\infty)$	R	R	R	R

Table 6
Inspection-maintenance policies under simulation.

Policy	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	S13
P4.over	$I(\infty)$	$I(\infty)$	$I(\infty)$	$I(2)$	M	M	R	R	R	R	R	R	R
P.Optimal	$I(\infty)$	$I(\infty)$	$I(\infty)$	$I(4)$	M	M	R	R	R	R	R	R	R
P4.under	$I(\infty)$	$I(\infty)$	$I(\infty)$	$I(6)$	M	M	R	R	R	R	R	R	R

Table 7
Monte Carlo simulation result.

Policy	# Failure	# Preventive maintenance/ replacement	Mean cost rate (,000 \$/month)	Normalized deviation of total cost (%)
P1	0.00	10.42	2.45	16.8
P2	0.01	8.95	2.29	20.5
P3	0.02	8.50	2.23	21.1
P4	0.32	3.15	2.20	32.5
P5	0.43	3.12	2.25	39.9
P7	0.46	3.07	2.30	40.9
P9	0.52	2.72	2.36	41.1
P4.over	0.10	5.92	2.12	23.6
P.Optimal	0.12	3.78	2.01	24.2
P4.under	0.25	3.16	2.18	30.7

can be seen that P4 is closed to the optimal policy P.Optimal, and hence only state 4 is further examined by letting its inspection to intervene at a different of 4 months as described in Table 6.

In Table 7, the mean overall cost rate under the optimal policy is the lowest as compared to the other policies. One can see that P.Optimal saves up to $(2.45-2.01)/2.45=18\%$ in cost of the least effective policy P1 and $(2.12-2.01)/2.12=5\%$ of the next most effective policy P4.over. The optimal policy is also superior to the policies P5, P7, P9 in failure protection as the mean number of failure under P.Optimal is only 0.12 which is about one forth of that for P5, P7 and P9 which are $0.43, 0.46, 0.52$, respectively. The normalized deviation of the total cost shows the percentage of fluctuation around the mean value. For policies with low maintenance threshold (P1, P2, P3), the deviation is as low as 16.8% since the system cost is kept stable by high rate of maintenance. For policies with high maintenance threshold (P5, P7, P9), the deviation is as high as 41.1% since failures may occurs highly random. For the optimal policy, it is 24.2 %, in between of them.

In view of the advantages and limitations of sequential and continuous inspection schemes, we propose a mixed inspection policy. The policy has the advantages of both inspection schemes, namely the ability to detect the change of system states timely and consistently by employing monitoring systems and the accuracy of the sequential inspection. In the optimal mixed inspection policy obtained, the time between inspections is well chosen so that the system degradation can be detected in time for proper maintenance actions.

5. Conclusion

In this work, we study the reliability of a multistate system under the condition of imperfect maintenance and continuous monitoring. The system has a state dependent degradation rate during its operation, and it also suffers shock failure which makes it fails immediately with a state dependent failure rate. At any time, the system may be suffering a random degradation which brings it to a lower performance state. Two inspection schemes are employed simultaneously, namely sequential and continuous monitoring. We present an iteration algorithm to determine the optimal inspection-maintenance policy. Monte Carlo simulation shows that the total cost under the optimal policies is smaller than that under the other maintenance policies.

The effectiveness of maintenance is highly important for an organization. Both the continuous monitoring and sequential inspection have their own advantages and they should be employed in combination. We proposed an analysis for such combination, and we show that the combination can be used optimally under the condition of imperfect maintenance, enhancing the system's reliability and maintenance effectiveness.

Appendix

Derivation of Eqs. (5) and (6)

$$X_\delta(i)=\int_0^{t_i} P_i(s(u)\in R(i))du+\sum_{j=1}^{N} P_i(s(t_i)=j)X_\delta(j) \tag{5}$$

$$Y_\delta(i)=A_i(t_i)+C^{SI}P_i(s(u)\neq N)+\sum_{j=1}^{N} P_i(s(t_i)=j)Y_\delta(j) \tag{6}$$

Eqs. (5) and (6) respectively represent the mean operating time and cost from the detected state i until it is recovered to as-good-as-new. The operating time and cost are derived under the decision to inspect the system after an operating duration of t_i. The waiting time t_i may be interrupted at a time $u<t_i$ if a change in operating range of states is detected or a failure occurs.

Denote $f_{ij}(t)$ as the probability density function of the probability $P_i(s(t)=j)$. The probability that the system does not fail and remain in the same operation range during the interval $(u, u+du)$ with $u<t_i$ is $P_i(s(u)\in R(i))$. While the probability that the system is interrupted by a change to state j out of the current operating rang is $f_{ij}(u)du$. Hence, the mean operating time if the system is left to operate until t_i is:

$$\begin{aligned} X_1 &= \int_0^{t_i}\left(P_i(s(u)\in R(i))+\sum_{j\notin R(i)} f_{ij}(u)X_\delta(j)\right)du \\ &= \int_0^{t_i} P_i(s(u)\in R(i))du+\sum_{j\notin R(i)} P_i(s(t_i)=j)X_\delta(j) \end{aligned} \tag{14}$$

If passing t_i, the system is re-inspected. Since there's no interruption during$(0, t_i)$, the additional mean time to renewal

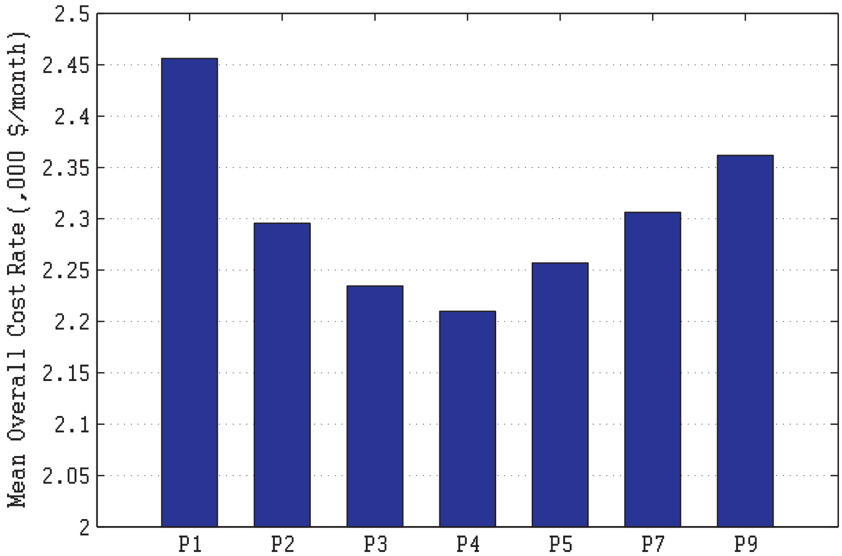

Fig. 3. Mean overall cost rate under different inspection-maintenance policies.

from t_i forward is:

$$X_2 = \sum_{j \in R(i)} P_i(s(t_i)=j)X_\delta(j) \tag{15}$$

Hence, the mean operating time to renewal is:

$$X_\delta(i) = X_1 + X_2 = \int_0^{t_i} P_i(s(u) \in R(i))du + \sum_{j \notin R(i)} P_i(s(t_i)=j)X_\delta(j)$$

$$+ \sum_{j \in R(i)} P_i(s(t_i)=j)X_\delta(j) \to X_\delta(i)$$

$$= \int_0^{t_i} P_i(s(u) \in R(i))du + \sum_{j=1}^{N} P_i(s(t_i)=j)X_\delta(j)$$

Similar derivation is applied for the Eq. (6). Assessing the system's mean cost in the interval $(u, u+du)$ with $u<t_i$, we have the system does not fail and remain in the same operation range result in a mean cost of $a_j du$ with the probability of $P_i(s(u) \in R(i))$. The system is interrupted by a change to state j out of the current operating rang result in a mean cost of $Y_\delta(j)$ with the probability of $f_{ij}(u)du$. Hence, we have

$$Y_1 = \int_0^{t_i} \left(\sum_{j \in R(i)} P_i(s(u)=j)a_j + \sum_{j \notin R(i)} f_{ij}(u) Y_\delta(j) \right) du = A_i(t_i)$$

$$+ \sum_{j \notin R(i)} P_i(s(t_i)=j)Y_\delta(j) \tag{16}$$

If passing t_i, the system is interfered by inspection. The additional mean cost to renewal from t_i forward is:

$$Y_2 = C^{SI} P_i(s(u) \neq N) + \sum_{j \in R(i)} P_i(s(t_i)=j)Y_\delta(j)$$

Hence the mean cost to renewal is:

$$Y_\delta(i) = Y_1 + Y_2 = A_i(t_i) + \sum_{j \notin R(i)} P_i(s(t_i)=j)Y_\delta(j) + C^{SI} P_i(s(u) \neq N)$$

$$+ \sum_{j \in R(i)} P_i(s(t_i)=j)Y_\delta(j) \to Y_{\delta(i)} = A_i(t_i) + + C^{SI} P_i(s(u) \neq N)$$

$$+ \sum_{j=1}^{N} P_i(s(t_i)=j)Y_\delta(j)$$

References

[1] Barlow RE, Proschan F, Hunter LC. Mathematical theory of reliability, vol. 17. Society for Industrial Mathematics; 1996.
[2] Grall A, Berenguer C, Dieulle L. A condition-based maintenance policy for stochastically deteriorating systems. Reliability Engineering & System Safety 2002;76:167–80.
[3] Lu S, Tu YC, Lu H. Predictive condition based maintenance for continuously deteriorating systems. Quality and Reliability Engineering International 2007;23:71–81.
[4] Ming Tan C, Raghavan N. A framework to practical predictive maintenance modeling for multi-state systems. Reliability Engineering & System Safety 2008;93:1138–50.
[5] Moustafa M, Maksoud E, Sadek S. Optimal major and minimal maintenance policies for deteriorating systems. Reliability Engineering & System Safety 2004;83:363–8.
[6] Barros A, Berenguer C, Grall A. A maintenance policy for two-unit parallel systems based on imperfect monitoring information. Reliability Engineering & System Safety 2006;91:131–6.
[7] Nakagawa T, Mizutani S, Chen M. A summary of periodic and random inspection policies. Reliability Engineering & System Safety 2010;95:906–11.
[8] Ohnishi M, Kawai H, Mine H. An optimal inspection and replacement policy for a deteriorating system. Journal of Applied Probability 1986;23:973–88.
[9] Lam C, Yeh R. Optimal maintenance-policies for deteriorating systems under various maintenance strategies. Reliability, IEEE Transactions on 1994;43: 423–30.
[10] Heinrich G, Jensen U. Optimal replacement rules based on different information levels. Naval Research Logistics (NRL) 1992;39:937–55.
[11] Nikiforov MBIV. Detection of abrupt changes: theory and application. Englewood Cliffs, NJ: PTR Prentice Hall; 1993.
[12] Simola K, Pulkkinen U. Models for non-destructive inspection data. Reliability Engineering & System Safety 1998;60:1–12.
[13] Sherwin D. Inspect or monitor? Engineering costs and production economics 1990;18:223–31.
[14] Lam CT, Yeh R. Comparison of sequential and continuous inspection strategies for deteriorating systems. Advances in Applied Probability 1994: 423–35.
[15] Pham H, Wang H. Imperfect maintenance. European Journal of Operational Research 1996;94:425–38.
[16] Kijima M. Some results for repairable systems with general repair. Journal of Applied Probability 1989;26:89–102.
[17] Kijima M, Morimura H, Suzuki Y. Periodical replacement problem without assuming minimal repair. European Journal of Operational Research 1988;37: 194–203.
[18] Zhou X, Xi L, Lee J. Reliability-centered predictive maintenance scheduling for a continuously monitored system subject to degradation. Reliability Engineering & System Safety 2007;92:530–4.
[19] Martorell S, Sanchez A, Serradell V. Age-dependent reliability model considering effects of maintenance and working conditions. Reliability Engineering & System Safety 1999;64:19–31.
[20] Sanchez A, Carlos S, Martorell S, Villanueva JF. Addressing imperfect maintenance modelling uncertainty in unavailability and cost based optimization. Reliability Engineering & System Safety 2009;94:22–32.
[21] Malik M. Reliable preventive maintenance policy. AIIE Transactions 1979;11:221–8.

[22] Chan JK, Shaw L. Modeling repairable systems with failure rates that depend on age and maintenance. Reliability, IEEE Transactions on 1993;42:566–71.
[23] Kallen M. Modelling imperfect maintenance and the reliability of complex systems using superposed renewal processes. Reliability Engineering & System Safety 2010.
[24] Liao GL, Chen YH, Sheu SH. Optimal economic production quantity policy for imperfect process with imperfect repair and maintenance. European Journal of Operational Research 2009;195:348–57.
[25] Nakagawa T, Yasui K. Optimum policies for a system with imperfect maintenance. Reliability, IEEE Transactions on 1987;36:631–3.
[26] Nakagawa T. Sequential imperfect preventive maintenance policies. Reliability, IEEE Transactions on 1988;37:295–8.
[27] Sheu SH, Lin YB, Liao GL. Optimum policies for a system with general imperfect maintenance. Reliability Engineering & System Safety 2006;91: 362–9.
[28] Wang H, Pham H. A quasi renewal process and its applications in imperfect maintenance. International journal of systems science 1996;27:1055–62.
[29] Brown M, Proschan F. Imperfect repair. Journal of Applied Probability 1983;20:851–9.
[30] Liu Y, Huang HZ. Optimal replacement policy for multi-state system under imperfect maintenance. Reliability, IEEE Transactions on 2010;59:483–95.
[31] Drenick R. The failure law of complex equipment. Journal of the Society for Industrial and Applied Mathematics 1960;8:680–90.
[32] Papoulis A, Pillai SU, Unnikrishna S. Probability, random variables, and stochastic processes, vol. 196. New York: McGraw-hill; 1965.